

Knowledge Distillation for Coral Bleaching Classification

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1 Introduction

Coral bleaching, a stress response where corals expel their symbiotic algae, threatens reef ecosystems worldwide and serves as an indicator of ocean health. Monitoring the extent and severity of bleaching events is critical for understanding ecosystem impacts and informing conservation efforts. Automated image classification using computer vision enables much easier coral reef monitoring, but deploying deep learning models onto edge devices in these resource-constrained field environments remains challenging.

While deep convolutional neural networks achieve high accuracy on image classification tasks, their computational requirements create substantial deployment barriers. Knowledge distillation [1] addresses this challenge by transferring knowledge from a large “teacher” model to a compact “student” model through soft probability distributions produced by temperature scaling, which carry richer information than hard labels alone. This technique essentially enables student models to mimic a teacher’s behavior (and therefore accuracy) on specific tasks such as binary image classification while being much less computationally intensive.

In this project, I compressed a ResNet50 [2] teacher model (23.5M parameters) into a MobileNetV3-Small [3] student (1.52M parameters)-a $15.5\times$ reduction-using the Kaggle Coral Bleaching Dataset (923 images). I systematically explored the temperature T and weighting parameter α hyperparameter space to identify optimal distillation configurations, implementing the distillation loss function manually with the T^2 gradient correction factor.

My findings were: (1) a knowledge distillation pipeline for coral bleaching classification; (2) a hyperparameter study revealing that conservative distillation ($T = 2.0$, $\alpha = 0.5$) outperforms more aggressive configurations; and (3) empirical evidence that distillation improves student performance beyond both the teacher (77.70% accuracy) and independently-trained student (78.42%), achieving 79.14% test accuracy with $15.5\times$ fewer parameters.

2 Methods

2.1 Dataset

I used the Kaggle Coral Bleaching Dataset [11], comprising 923 RGB images labeled as healthy (438 samples, 47.5%) or bleached (485 samples, 52.5%). Images have variable resolutions ranging from 200×179 to 300×300 pixels. I split the dataset into training (645 images), validation (139 images), and test (139 images) sets using stratified sampling with random seed 42. Training images undergo data augmentation: random resized crop to 224×224 pixels, random horizontal flip ($p=0.5$), color jitter (brightness/contrast/saturation ± 0.2), and normalization using ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). Validation and test images are resized to 256×256 , center-cropped to 224×224 , and normalized identically.

2.2 Model Architectures

Teacher Model. I used ResNet50 [2] pre-trained on ImageNet (23.5M parameters), modifying the final layer for binary classification and adding dropout ($p = 0.5$) for regularization.

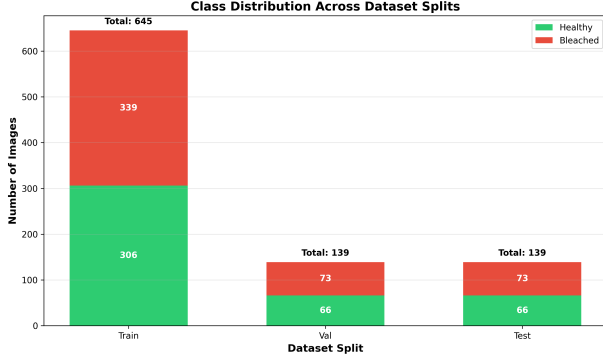


Figure 1: Class distribution across train, validation, and test splits. The dataset is reasonably balanced with 438 healthy and 485 bleached coral images.

Student Model. I employed MobileNetV3-Small [3], a lightweight architecture with 1.52M parameters ($15.5\times$ reduction), modifying the final layer for binary classification.

2.3 Training Procedures

2.3.1 Manual Distillation Implementation

To demonstrate understanding of the underlying mathematics, I manually implemented the distillation mechanism rather than using high-level PyTorch functions. Temperature-scaled softmax is computed as:

$$p_i^{(T)} = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (1)$$

where z_i are logits and T is temperature. Higher T produces softer probability distributions. The KL divergence between teacher and student soft targets is:

$$\mathcal{L}_{\text{KL}} = \sum_i p_i^{\text{teacher}} \log \left(\frac{p_i^{\text{teacher}}}{p_i^{\text{student}}} \right) \quad (2)$$

The T^2 factor in Equation 3 ensures proper gradient magnitudes during backpropagation [1].

2.3.2 Training Configuration

I trained all models using Adam optimizer (learning rate 1×10^{-4}) with early stopping (pa-

tience 10 epochs). The teacher and student baseline used standard cross-entropy loss. For knowledge distillation training, I combined hard-label cross-entropy with soft-target KL divergence from the frozen teacher using the combined loss (Equation 3):

$$\mathcal{L}_{\text{KD}} = \alpha \cdot \mathcal{L}_{\text{hard}} + (1 - \alpha) \cdot T^2 \cdot \mathcal{L}_{\text{soft}} \quad (3)$$

where $\mathcal{L}_{\text{hard}}$ is cross-entropy with true labels, $\mathcal{L}_{\text{soft}}$ is KL divergence between teacher and student soft targets, T is temperature, and α is the weighting factor. The T^2 scaling ensures proper gradient magnitudes [1]. All experiments used random seed 42 for reproducibility and were trained on Google Colab T4 GPUs.

2.4 Hyperparameter Exploration

I explored temperature $T \in \{2.0, 4.0, 8.0\}$ and weighting factor $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ using four strategic configurations: Conservative ($T = 2.0$, $\alpha = 0.5$), Label-focused ($T = 4.0$, $\alpha = 0.3$), Default ($T = 4.0$, $\alpha = 0.7$), and Aggressive ($T = 8.0$, $\alpha = 0.9$). All configurations used identical training settings for fair comparison.

3 Results

Table 1 summarizes performance metrics for all six models, and Figure 2 provides a comprehensive visual overview.

3.1 Overall Performance Comparison

Table 1 and Figure 2A presented the performance comparison across all models. The ResNet50 teacher achieved 77.70% test accuracy with 23.5M parameters. The MobileNetV3-Small student baseline achieved 78.42% test accuracy with 1.52M parameters ($15.5\times$ reduction), outperforming the teacher by +0.72% despite dramatically smaller capacity. The student exhibited 74.2% precision and 90.4% recall on healthy corals versus 86.0% precision and 65.2% recall on bleached corals. Model checkpoints reduced from 270 MB to 18 MB, enabling deployment on edge devices (Figure 2B).

Table 1: Performance comparison of teacher, student baseline, and distilled students with different hyperparameters. Best results in bold.

Model	Test Acc (%)	Precision (%)	Recall (%)	F1-Score (%)	Parameters
Teacher (ResNet50)	77.70	77.68	77.75	77.68	23,512,130
Student Baseline (MobileNetV3)	78.42	80.08	77.78	77.81	1,519,906
KD ($T = 2.0, \alpha = 0.5$)	79.14	79.75	79.48	79.12	1,519,906
KD ($T = 4.0, \alpha = 0.3$)	78.42	78.48	78.22	78.28	1,519,906
KD ($T = 4.0, \alpha = 0.7$)	76.26	76.24	76.31	76.24	1,519,906
KD ($T = 8.0, \alpha = 0.9$)	78.42	79.02	78.00	78.09	1,519,906

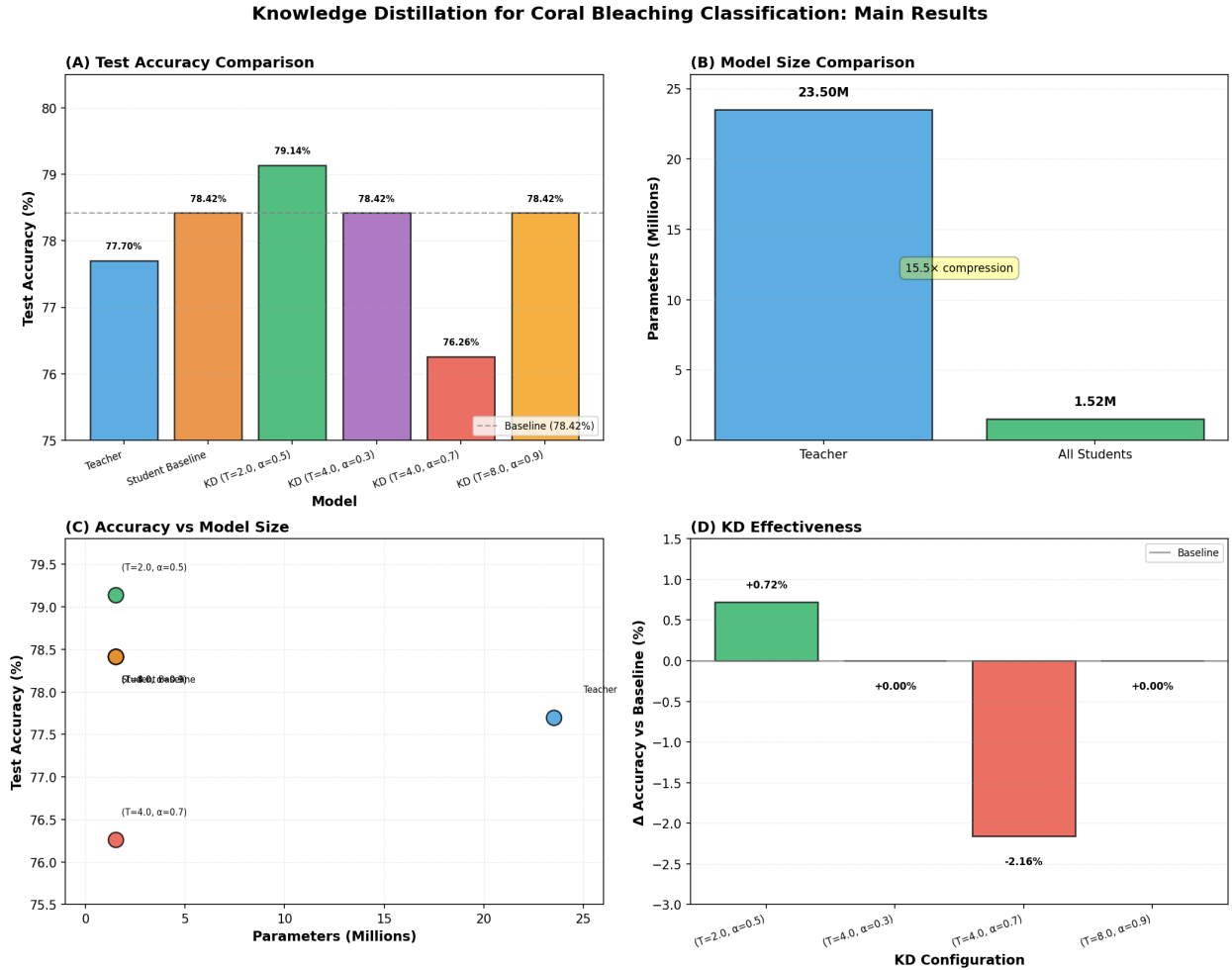


Figure 2: Comprehensive results summary. (A) Test accuracy comparison across all models. (B) Model size comparison showing 15.5 $\times$ compression. (C) Accuracy vs efficiency tradeoff (Pareto frontier). (D) KD effectiveness showing improvement over baseline for different hyperparameter configurations.

3.2 Knowledge Distillation Results

I evaluated four knowledge distillation configurations spanning temperature $T \in \{2.0, 4.0, 8.0\}$ and weighting parameter $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$. The best configuration ($T = 2.0$, $\alpha = 0.5$) achieved 79.14% test accuracy, improving +0.72% over the student baseline and +1.44% over the teacher. With 79.75% precision and 79.48% recall, this model achieved balanced per-class performance. The label-focused configuration ($T = 4.0$, $\alpha = 0.3$) matched baseline at 78.42% accuracy, while the aggressive configuration ($T = 8.0$, $\alpha = 0.9$) also achieved 78.42%. Notably, the default literature configuration ($T = 4.0$, $\alpha = 0.7$) underperformed at 76.26% accuracy—a -2.16% drop from baseline—demonstrating that standard hyperparameters are not universally optimal. Performance varied from -2.16% to +0.72% relative to baseline across configurations.

3.3 Error Pattern Analysis

Figure 3 presented confusion matrices for all six models. The teacher correctly classified 56 of 73 healthy corals (76.7% recall) and 52 of 66 bleached corals (78.8% recall), with 17 false negatives and 14 false positives. The student baseline correctly classified 66 of 73 healthy corals (90.4% recall) but only 43 of 66 bleached corals (65.2% recall), producing 7 false negatives and 23 false positives. The best distilled model ($T = 2.0$, $\alpha = 0.5$) achieved balanced performance with approximately 79% recall on both classes, effectively bridging the complementary biases of teacher and student. The underperforming configuration ($T = 4.0$, $\alpha = 0.7$) degraded performance on both classes, while other configurations maintained baseline error distributions.

3.4 Hyperparameter Sensitivity

Temperature $T \in \{2.0, 4.0, 8.0\}$ revealed a trend favoring lower temperatures: $T = 2.0$ achieved 79.14% accuracy, while $T = 4.0$ produced mixed results (76.26–78.42%) and $T = 8.0$

matched baseline at 78.42%. Sharper distributions preserved stronger teacher guidance for binary classification. Weighting parameter $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ showed balanced weighting as optimal: $\alpha = 0.5$ achieved 79.14%, while extreme values ($\alpha = 0.3$, $\alpha = 0.9$) matched baseline at 78.42%, and $\alpha = 0.7$ produced the worst result at 76.26%. The best combination ($T = 2.0$, $\alpha = 0.5$) paired conservative temperature with balanced weighting, while the standard literature default ($T = 4.0$, $\alpha = 0.7$) underperformed. Performance variation from -2.16% to +0.72% revealed high sensitivity to hyperparameter selection.

4 Discussion

4.1 Why Did Student Baseline Outperform Teacher?

The student’s superior performance (+0.72% despite $15.5\times$ fewer parameters) suggests ResNet50 is overparameterized for binary coral classification. With only 645 training images, the 23.5M-parameter teacher shows a larger generalization gap (5.3%: 83.0% validation $\rightarrow$ 77.70% test) compared to the student’s smaller gap (3.6%: 82.01% $\rightarrow$ 78.42%), consistent with overfitting. The student’s constrained capacity acts as implicit regularization, forcing generalizable feature learning. This shows that matching model capacity to dataset scale matters more than deploying the largest architecture.

4.2 Why Did $T = 2.0$, $\alpha = 0.5$ Work Best?

Lower temperature ($T = 2.0$) produces sharper distributions that preserve discriminative information for binary classification, while higher temperatures ($T = 8.0$) may obscure the distinction between healthy and bleached corals. This contrasts with multi-class ImageNet defaults where softer distributions reveal inter-class similarities. Balanced weighting ($\alpha = 0.5$) effectively leverages both ground truth and teacher knowledge, while extreme values fail: over-emphasizing hard labels ($\alpha = 0.3$) ne-

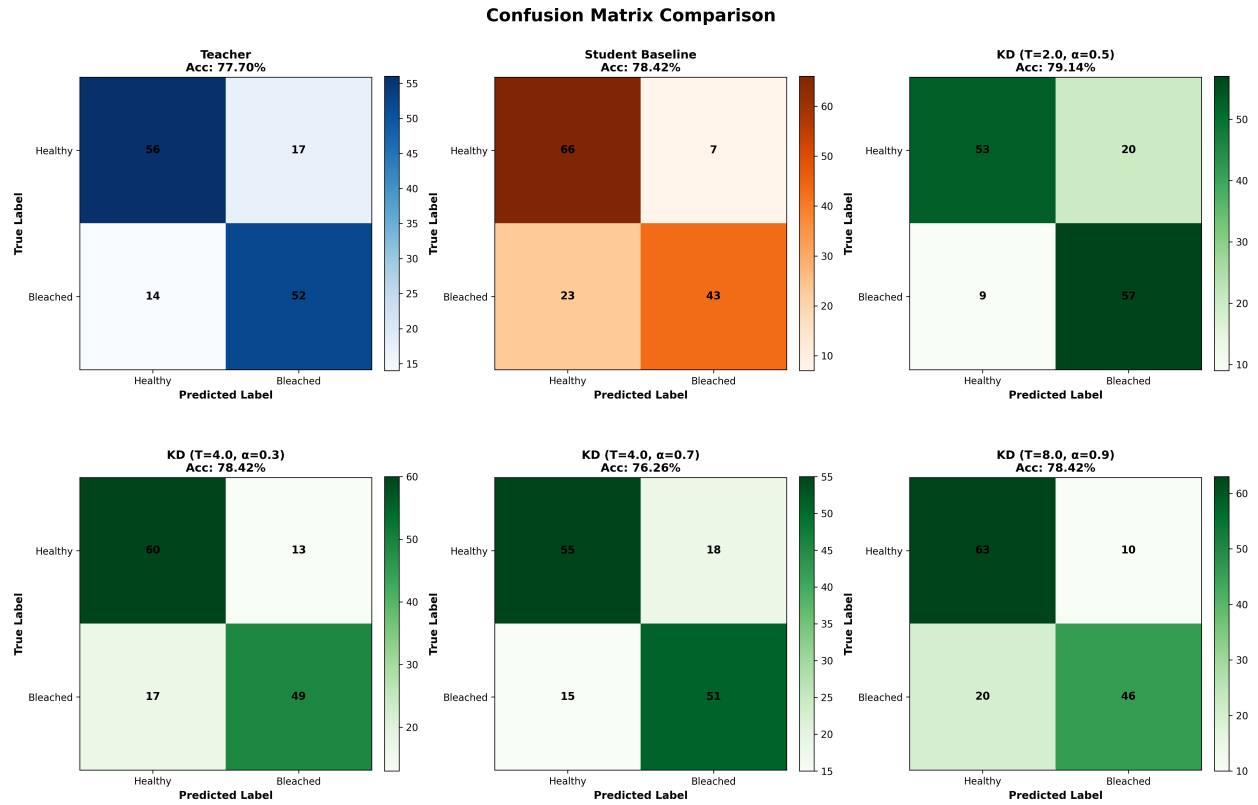


Figure 3: Confusion matrices for all six models. The 2×3 grid showed error patterns across teacher, baseline, and four distilled students. Best KD ($T = 2.0, \alpha = 0.5$) showed most balanced predictions across both classes.

glects teacher guidance, while over-relying on the teacher ($\alpha = 0.7$, $\alpha = 0.9$) propagates its weaknesses. The conservative configuration creates optimal knowledge transfer, while standard defaults ($T = 4.0$, $\alpha = 0.7$) amplify teacher errors.

4.3 Confidence Calibration

The best distilled model ($T = 2.0$, $\alpha = 0.5$) exhibits a confidence gap of 0.153 between correct and incorrect predictions, compared to 0.118 for the student baseline-30% higher (Figure 4). This improvement stems from the teacher’s soft probability distributions, which convey uncertainty estimates absent from hard labels. Well-calibrated models enable field deployment where high-confidence predictions proceed autonomously while low-confidence cases trigger expert review.

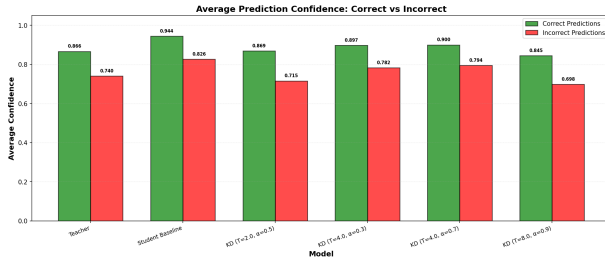


Figure 4: Average prediction confidence for correct vs incorrect predictions across all models. Best KD ($T = 2.0$, $\alpha = 0.5$) shows highest confidence gap (0.153), indicating better calibration.

4.4 Limitations and Future Work

Limitations:

- Small dataset (923 images) may limit generalization to other coral species, regions, and imaging conditions
- Single teacher-student architecture pair (ResNet50 $\rightarrow$ MobileNetV3-Small)
- Strategic sampling of four configurations, not exhaustive hyperparameter search
- No deployment benchmarks on actual edge hardware (latency, energy consumption)

- Single training runs without variance estimates or confidence intervals

Future Work:

- Combine KD with quantization and pruning for greater compression (50-100 $\times$)
- Evaluate on actual edge hardware (Raspberry Pi, Jetson, underwater vehicles)
- Cross-domain validation on diverse coral datasets
- Extend to multi-class bleaching severity prediction
- Exhaustive hyperparameter search or Bayesian optimization

4.5 Summary

Knowledge distillation successfully compressed ResNet50 (23.5M parameters) to MobileNetV3-Small (1.52M parameters)-a 15.5 $\times$ reduction-while improving accuracy. My best configuration ($T = 2.0$, $\alpha = 0.5$) achieved 79.14% test accuracy (+0.72% over baseline, +1.44% over teacher) with superior calibration. Conservative hyperparameters outperformed literature defaults by 2.88%, demonstrating that task-specific tuning is essential. This enabled practical deployment on edge devices for real-time coral reef monitoring without cloud infrastructure.

AI Acknowledgment

The course syllabus states we should acknowledge the use of AI in this project. I used AI for the following tasks in this project:

- **Research:** I used AI (Deep Research mostly) to browse the web for sources to read on these topics.
- **Code Debugging:** This codebase is quite large, and some of the concepts were unfamiliar to me implementation-wise. I used AI to help debug when I ran into issues I did not know how/have the time to solve.
- **LaTeX Formatting:** I used AI for much of the graph/table formatting in this report.
- **Proofreading:** I used AI to give me a thorough review of this report.

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